

Pixels and Filters

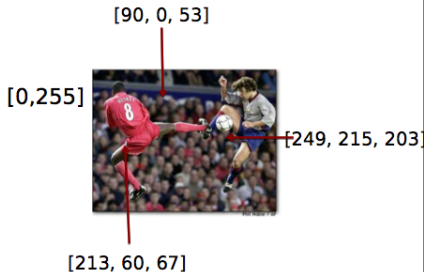
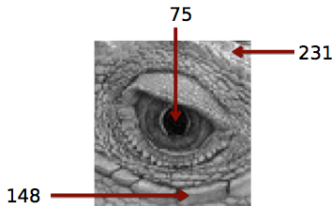
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- Images as functions
- Point Operators
- Linear System (filters)
- Convolution and Correlation

Images as function

- An image contains discrete number of pixels
 - Pixel value:
 - "grayscale" (or "intensity"): $[0, 255]$
 - "color" - RGB = $[R, G, B]$; Lab = $[L, a, b]$; HSV = $[H, S, V]$



Images as function

- An Image as a function f from R^2 to R^M :
 - $f(x, y)$ gives the intensity at position (x, y)
 - Defined over a rectangle, with a finite range
 $f : [a, b] \times [c, d] \longrightarrow [0, 255]$
- a color image in RGB space: $f(x, y) = \begin{pmatrix} r(x, y) \\ g(x, y) \\ b(x, y) \end{pmatrix}$

Images as discrete functions

- Images are usually digital (discrete)
- Represented as matrix of integer values

pixel

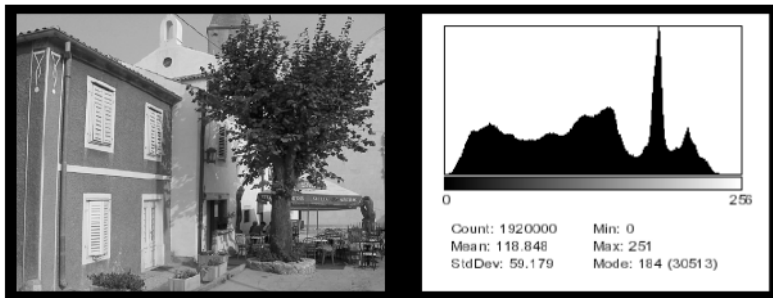


j	62	79	23	119	120	05	4	0
i	10	10	9	62	12	78	34	0
	10	58	197	46	46	0	0	48
	176	135	5	188	191	68	0	49
	2	1	1	29	26	37	0	77
	0	89	144	147	187	102	62	208
	255	252	0	166	123	62	0	31
	166	63	127	17	1	0	99	30

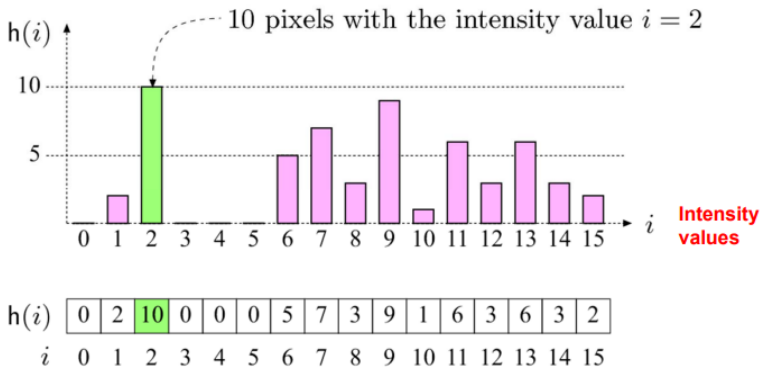
Histogram and Point Operations

Histogram

- Histograms plots how many times (frequency) each intensity value in image occurs
- Example
 - Image (left) has 256 distinct gray levels (8 bits)
 - Histogram (right) shows frequency (how many times) each gray level occurs



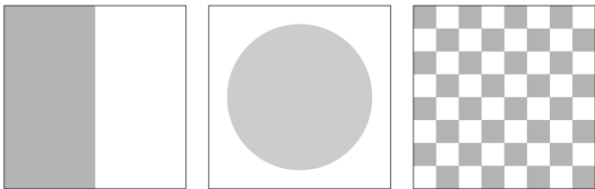
Histograms



- E.g. $K = 16$, 10 pixels have intensity value = 2
- Histograms: only statistical information
- No indication of location of pixels

Histograms

- Different images can have same histogram
- 3 images below have same histogram



- Half of pixels are gray, half are white
 - Same histogram = same statistics
 - Distribution of intensities could be different
- Can we reconstruct image from histogram? No!

- Histograms help detect image acquisition issues
- Problems with image can be identified on histogram
 - Over and under exposure
 - Brightness
 - Contrast
 - Dynamic Range
- Point operations can be used to alter histogram.
 - Addition
 - Multiplication
 - Exp and Log
 - Intensity Windowing (Contrast Modification)

Image Brightness

- Brightness of a grayscale image is the average intensity of all pixels in image

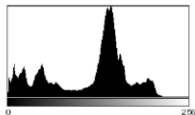
$$B(I) = \frac{1}{wh} \sum_{v=1}^h \sum_{u=1}^w I(u, v)$$

2. Divide by total number of pixels

1. Sum up all pixel intensities

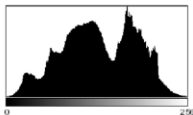
Detecting Bad Exposure using Histograms

- Exposure? Are intensity values spread (**good**) out or bunched up (**bad**)



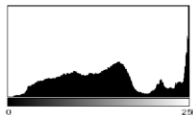
(a)

Underexposed



(b)

**Properly
Exposed**



(c)

Overexposed

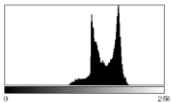
Image

Histogram

- The contrast of a grayscale image indicates how easily objects in the image can be distinguished
- **High contrast** image: many distinct intensity values
- **Low contrast**: image uses few intensity values

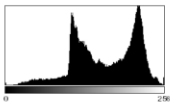
Histograms and Contrast

- **Good Contrast?** Widely spread intensity values + large difference between min and max intensity values



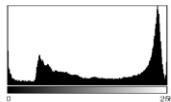
(a)

Low contrast



(b)

Normal contrast



(c)

High contrast

Image

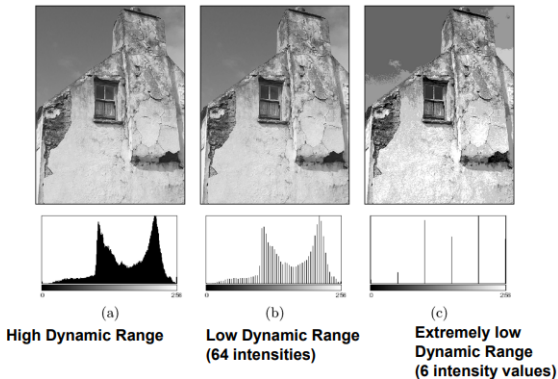
Histogram

Contrast Equation

- Many different equations for contrast exist
- Michalson's equation for contrast
$$C_M(I) = \frac{\max(I) - \min(I)}{\max(I) + \min(I)}$$
- These equations work well for simple images with 2 luminances (i.e. uniform foreground and background)
- Does not work well for complex scenes with many luminances or if min and max intensities are small

Histograms and Dynamic Range

- Dynamic Range: Number of distinct pixels in image

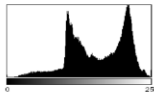


- Difficult to increase image dynamic range
- High dynamic range means very bright and very dark parts in a single image (many distinct values)

Detecting Image Defects using Histograms

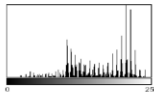
- No “best” histogram shape, depends on application
- Image defects
 - Saturation: scene illumination values outside the sensor’s range are set to its min or max values => results in spike at ends of histogram
 - Effect of Image Compression: Example in GIF compression, dynamic range is reduced to only few intensities

Original Image



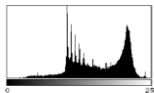
(a)

Original Histogram



(b)

Histogram after GIF conversion



(c)

Fix? Scaling image by 50% and
Interpolating values recreates
some lost colors
But GIF artifacts still visible

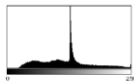
Color Image Histograms

Two types

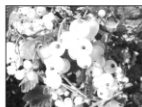
- Intensity histogram:
 - Convert color image to gray scale
 - Display histogram of gray scale
- Individual Color Channel Histograms: 3 histograms (R,G,B)
 - 3 histograms (R,G,B)



(a)



(b) h_{Lum}



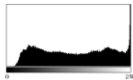
(c) R



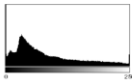
(d) G



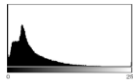
(e) B



(f) h_R



(g) h_G



(h) h_B

Color Image Histograms

- Both types of histograms provide useful information about lighting, contrast, dynamic range and saturation effects
- No information about the actual color distribution!
- Images with totally different RGB colors can have same R, G and B histograms
- Solution to this ambiguity is the Combined Color Histogram.

Cumulative Histogram

- Useful for certain operations (e.g. histogram equalization)
- Analogous to the Cumulative Density Function (CDF)
- Definition:

$$H(i) = \sum_{j=0}^i h(j) \text{ for } 0 \leq i < K \quad (1)$$

- Recursive definition

$$H(i) = \begin{cases} h(0) & \text{for } i = 0 \\ H(i-1) + h(i) & \text{for } 0 < i < K \end{cases} \quad (2)$$

- Monotonically increasing

$$H(K-1) = \sum_{j=0}^{K-1} h(j) = M \cdot N \quad (3)$$

Point Operations

- Point operations changes a pixel's intensity value according to some function (don't care about pixel's neighbor)
- Also called a **homogeneous operation**
- New pixel intensity **depends on**
 - Pixel's previous intensity $I(u, v)$
 - Mapping function $f()$
- **Does not depend on**
 - Pixel's location (u, v)
 - Intensities of neighboring pixels

Some Point Operations

- Addition
- Multiplication
- Exp and Log
- Intensity Windowing (Contrast Modification)